Question Paper

Exam Date & Time: 22-Nov-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: CSS 2102/CSS_2102 DATA COMMUNICATION COMPUTER NETWORKS
(SPL: COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE / CYBER SECURITY / DATA SCIENCE / INFORMATION TECHNOLOGY)

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) Compare Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK) in terms of how each modifies the carrier signal, their bandwidth efficiency, and susceptibility to noise. Analyse which technique is most suitable for transmission over a noisy telephone channel and justify your choice. (5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

1B) Assume that the sender's window size is 3. If we have to send 10 frames in total, and the channel of transmission is such that every 5th transmission is lost (but no ACKs), how many data transmissions (including both original transmissions and any necessary retransmissions) does the sender make in total if:- (3)
i. The go-back-N protocol is followed
ii. The selective repeat protocol is followed
Also, calculate the ratio of the effective number of frames sent to the total number of frames sent in each of the above cases.

Course Outcome: "CO2" / Cognitive Level: "Application"

1C) A system has an n-layer protocol hierarchy. Applications generate messages of length M bytes. At each of the layers, an h-byte header is added. What fraction of the network bandwidth is filled with headers? (2)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

2A) Suppose an ISP owns the block of addresses of the form 128.119.40.96/26. Suppose it wants to create four subnets from this block, with each block having the same number of IP addresses. What are the prefixes (of form a.b.c.d/x) for the four subnets? Draw the network diagram for the same. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

2B) Aggregate the following set of four /24 IP network addresses to the highest degree possible. (3)
212.56.146.0/24
212.56.147.0/24
212.56.148.0/24
212.56.149.0/24.
Find the CIDR aggregation.

Course Outcome: "CO3" / Cognitive Level: "Application"

2C) Consider the following routing table at an IP router: (2)

Network No	Net Mask	Next Hop
128.96.170.0	255.255.254.0	Interface 0
128.96.168.0	255.255.254.0	Interface 1
128.96.166.0	255.255.254.0	R2
128.96.164.0	255.255.252.0	R3
0.0.0.0	Default	R4

Show the forwarding process of the following packets at the router with the following destination addresses:
(a) 128.96.171.92
(b) 128.96.167.151.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

3A) A transport layer segment of 3500 bytes is sent from Host P to Host Q through two networks connected by Router R2. The first network (from Host P to Router R2) supports a maximum transmission unit (MTU) of 1000 bytes, while the second network (from Router R2 to Host Q) supports an MTU of 800 bytes. Assume that each IP header is 20 bytes long. If all fragments successfully reach the destination, calculate how fragmentation occurs in each network and specify for every fragment the data size, offset value, and More Fragments (MF) flag. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

3B) Consider the network scenario shown in the figure below. Host PC0 with IP address 192.168.10.9/24 intends to send an IP datagram to Host PC3 with IP address 192.168.20.11/24, but PC0 does not have the MAC address of PC3. Both hosts are located on different Ethernet networks connected through Router0. The G0/0 interface of Router0, which connects to the network of PC0, has IP address 192.168.10.1 and MAC address 57:3E:32:42:AC:AD, while the G0/1 interface, connected to the network of PC3, has IP address 192.168.20.1 and MAC address AC:AE:32:45:73:42. Assume that PC0 has the MAC address 10:90:21:C4:69:A7 and PC3 has the MAC address B2:22:F4:59:83:AB. (3)
Show the ARP request packet sent and Reply packets received by Host PC0. Also show the Ethernet frames with suitable fields.

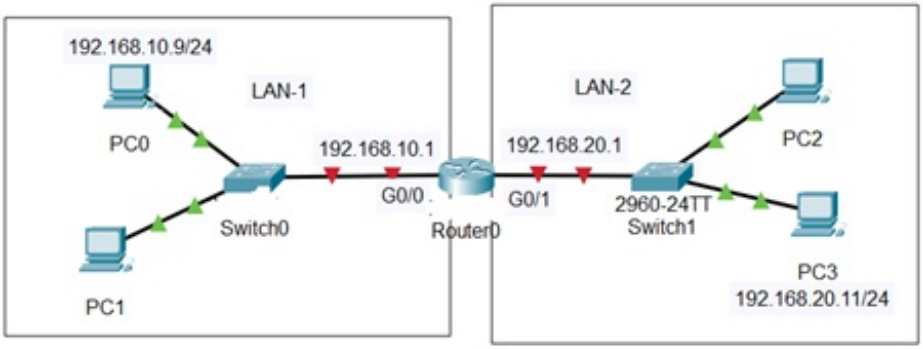


Figure. 3(B)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

3C) For the given binary data 101000101010, draw the following line coding scheme. (Assume previous pulse was negative) (2)
i. Differential Manchester
ii.Bipolar AMI

Course Outcome: "CO1" / Cognitive Level: "Application"

4A) A network engineer at a logistics company is optimizing data packet delivery across a wide-area network (WAN) connecting multiple regional data centres. The network is modelled as a weighted graph, where each node represents a router located in a city. Each edge weight represents the transmission delay (in milliseconds) between routers. (5)

The network topology is as follows:
Router A → Router B : 4 ms
Router A → Router C : 2 ms
Router C → Router B : 3 ms
Router B → Router D : 3 ms
Router C → Router D : 2 ms
Router D → Router E : 3 ms
Router C → Router E : 6 ms
Apply Dijkstra's algorithm to determine the minimum latency path from Router A to Router E. Show all steps clearly. Also construct the routing table for Router A.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

4B) During a TCP data transfer, Host A sends three consecutive segments to Host B. The first segment (1001-2000) and the third segment (3001-4000) reach the receiver, but the second segment (2001-3000) is delayed in the network. Later, the delayed segment arrives, followed by a duplicate of the third segment due to retransmission. Based on this scenario, describe how Host B generates acknowledgments for each event according to TCP acknowledgment rules when: (3)
(a) An out-of-order segment is received,
(b) Missing segment is received
(c) A duplicate segment is detected.

Course Outcome: "CO4" / Cognitive Level: "Application"

4C) Calculate the total time required to transmit 1,000,000 bits using the Stop-and-Wait ARQ protocol when each frame carries 2000 bits, the sender and receiver are 5000 km apart, and the propagation speed is 2 × 10^8 m/s. Ignore transmission, waiting, and processing delays, and assume no frames are lost or damaged (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

5A) Using the concept of the three-way handshake, illustrate step-by-step how a TCP connection is established between a client and a server. Identify the flags used in each segment and explain their purpose. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

5B) Describe the processes of recursive resolution and iterative resolution in DNS, using appropriate diagrams to illustrate each method. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

5C) An application sends 2048 bytes of data using UDP. The IP layer adds a 20-byte header, and the datalink layer adds a 14-byte header and 4-byte trailer. Find the total frame size transmitted on the network. (UDP header is 8 byte). (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
END TERM EXAM-ODD SEMESTER 2025 MIT MANIPAL	CSS 2102:DATA COMMUNICATION AND COMPUTER NETWORKS

#	Course Outcome	Description	Marks

1	1	CO 1		7
2	2	CO 2		7
3	3	CO 3		18
4	4	CO 4		10
5	5	CO 5		8
			Total	**50**

* NA - CO Not Selected

Event Name	Subject
END TERM EXAM-ODD SEMESTER 2025 MIT MANIPAL	CSS 2102:DATA COMMUNICATION AND COMPUTER NETWORKS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	9	28(56%)
4	Analysis	5	17(34%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)